

Student Performance Prediction System

Project Report

Submitted by

Ekta Zope

Project

Student Performance Prediction System

Technology

Python & Machine Learning

1. Introduction

The Student Performance Prediction System is a beginner-level Machine Learning project designed to analyze student-related data and predict their academic performance.

The project uses factors such as study hours, attendance, previous marks, assignment scores, sleep hours, and internet access to predict whether a student's performance is **Low, Medium, or High**.

The project was developed using Python and basic Machine Learning libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn.

2 Project Objective

The main objective of this project is to build a basic Machine Learning system that can analyze student-related information and predict their performance category.

The project aims to:

- Analyze student academic and lifestyle data.
- Clean and prepare the dataset.
- Identify patterns in the data using visualizations.
- Select relevant features for prediction.
- Train a Machine Learning model.
- Evaluate the model's predictions.
- Predict the performance of a new student.

3. Technologies Used

- **Python** – Programming language used for the project.
- **Pandas** – Used for data loading, cleaning, and analysis.
- **NumPy** – Used for numerical operations.
- **Matplotlib** – Used to create graphs and visualize data.
- **Scikit-learn** – Used for Machine Learning, data splitting, model training, and evaluation.
- **Jupyter Notebook** – Used to write and execute the Python code.

4. Dataset Description

The dataset contains information about students and their academic and lifestyle-related factors.

The dataset includes the following columns:

Column	Description
Study_Hours	Number of hours a student studies
Attendance	Student attendance percentage
Previous_Marks	Marks obtained previously
Assignment_Score	Assignment score
Sleep_Hours	Average hours of sleep
Internet_Access	Whether the student has internet access
Performance	Students performance category

5. Project Workflow

The project follows these steps:

**Data Collection → Data Cleaning → Data Analysis → Feature Selection
→ Train-Test Split → Model Training → Model Evaluation → Performance
Prediction**

6.Data Cleaning

Data cleaning was performed before training the Machine Learning model. The dataset was checked for missing values and duplicate records.

The following checks were performed:

- Checked for missing values.
- Checked for duplicate records.
- Checked the data types of the columns..
- Converted text-based values such as Internet Access and Performance into numerical values for Machine Learning.

7.Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the dataset and identify patterns between different student-related factors.

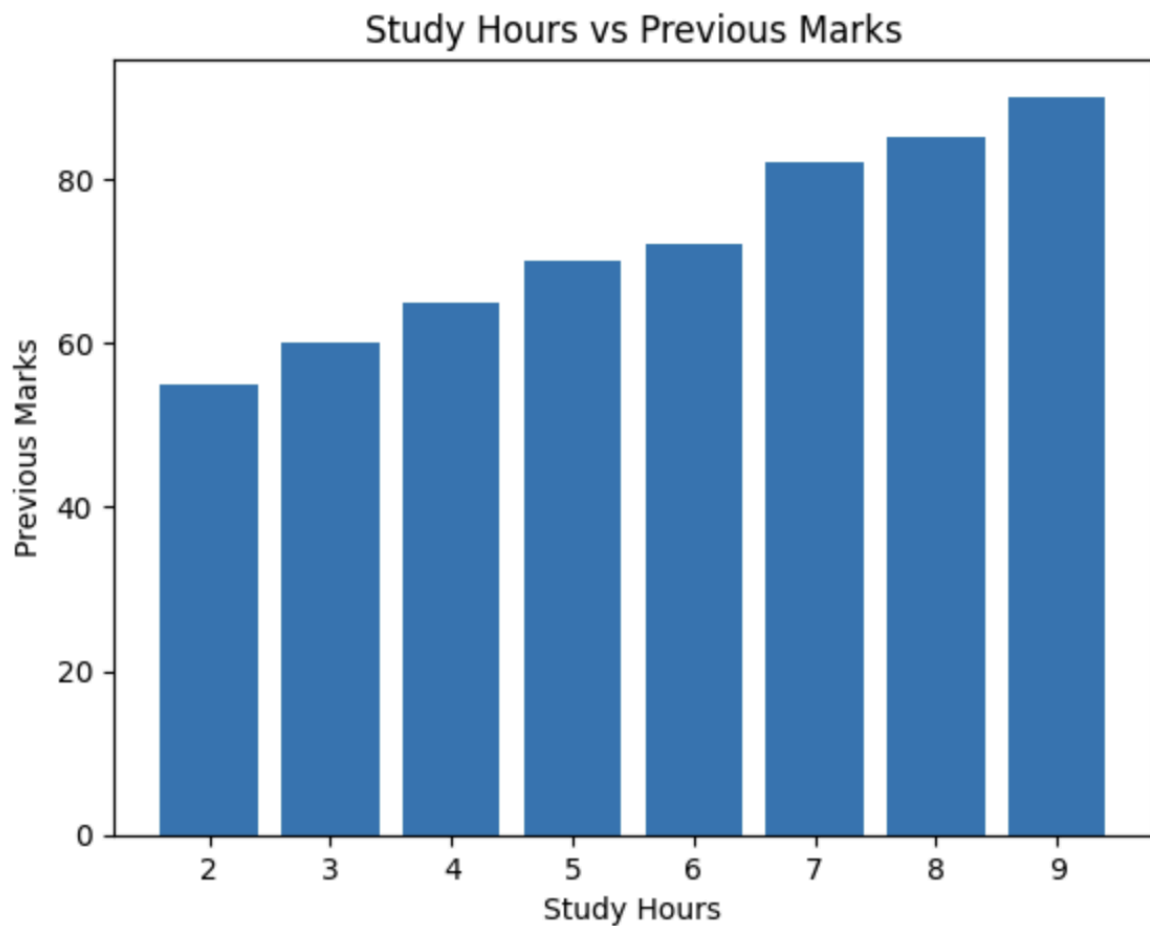
The following visualizations were created:

- Study Hours vs Previous Marks
- Attendance vs Assignment Score
- Student Performance Distribution

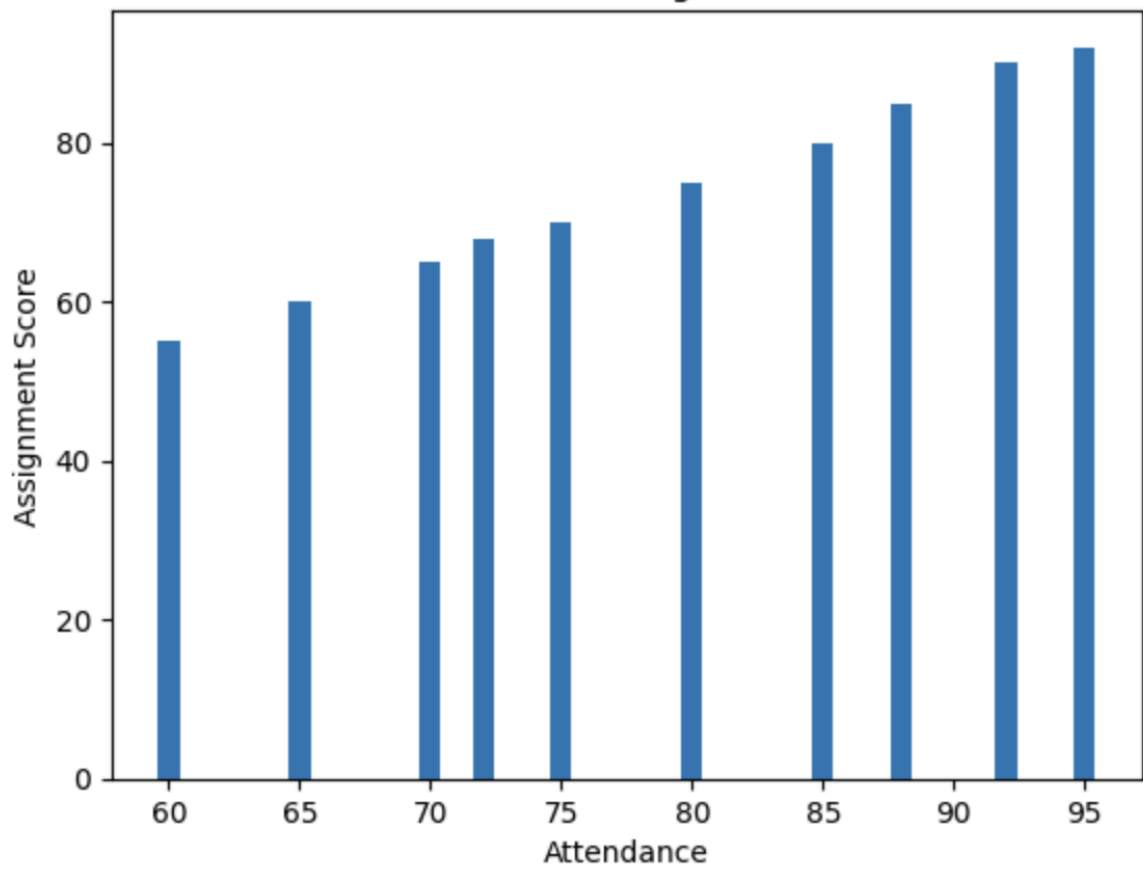
The analysis showed that students with higher study hours generally had higher previous marks in this dataset. Students

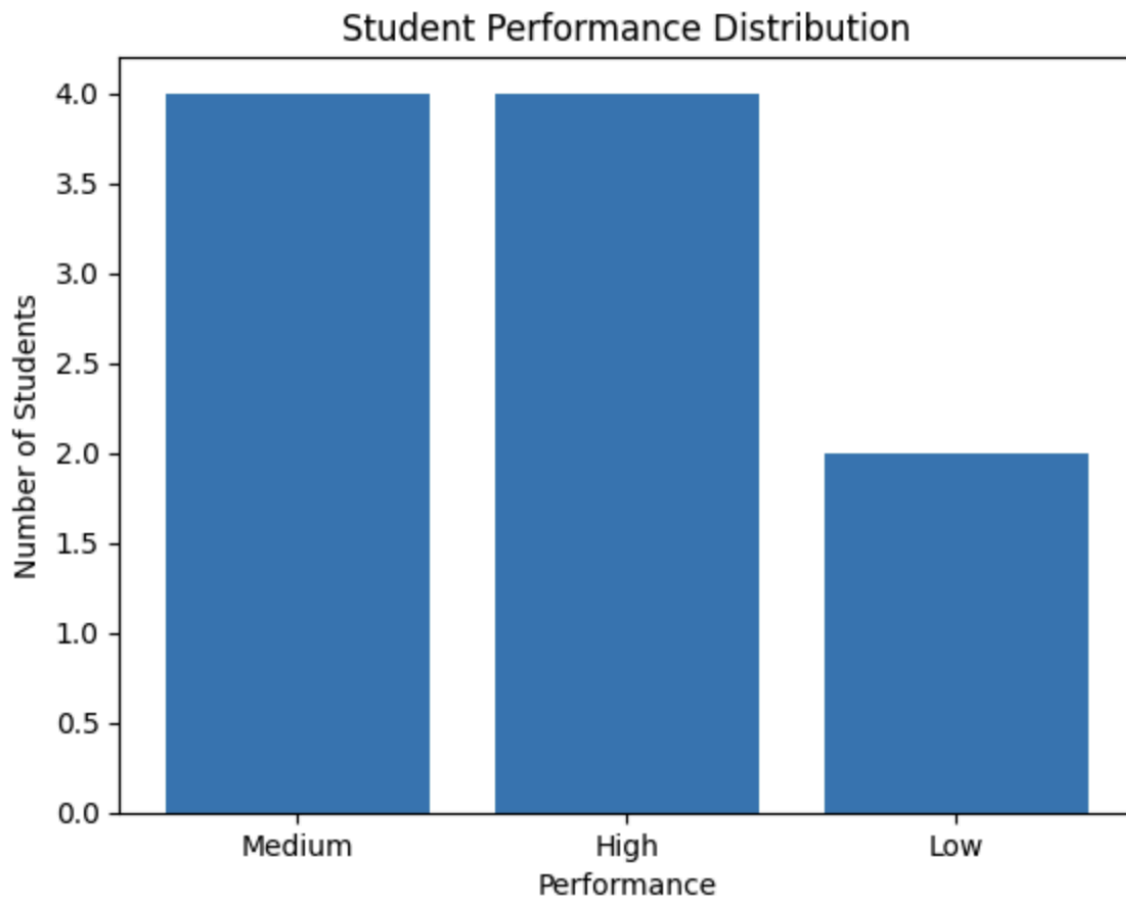
with higher attendance also generally had higher assignment scores.

The performance distribution contained three categories: **Low**, **Medium**, and **High**.



Attendance vs Assignment Score





8.Feature Selection

The following features were selected to predict student performance:

- Study Hours
- Attendance
- Previous Marks
- Assignment Score
- Sleep Hours
- Internet Access

The target variable is **Performance**.

These features were used as inputs for training the Machine Learning model.

9. Train-Test Split

The dataset was divided into two parts: training data and testing data.

- **80% of the data** was used for training the Machine Learning model.
- **20% of the data** was used for testing the model.

The training data was used to teach the model, while the testing data was used to check how well the model performed on unseen data.

10. Model Training

A **Decision Tree Classifier** was used as the Machine Learning model.

The model was trained using the training dataset. The selected student features were provided as inputs, and **Performance** was used as the target variable.

An improved Decision Tree model was also created by limiting the maximum tree depth to **3**. This was done to make the model simpler and reduce the possibility of overfitting.

Model trained successfully!

11. Model Evaluation

The trained model was evaluated using the **accuracy score**. The predictions made by the model were compared with the actual performance categories in the testing dataset.

A confusion matrix and an evaluation table were also used to compare the actual and predicted values.

The model was also tested with a new student's information to generate a performance prediction.

Actual: [2 1]
Predicted: [2 1]

Accuracy: 1.0

12. Prediction Result

The trained model was used to predict the performance of a new student.

The sample student had the following information:

- Study Hours: 5
- Attendance: 80%
- Previous Marks: 70
- Assignment Score: 75

- Sleep Hours: 7
- Internet Access: Yes

The model predicted the student's performance as:

Predicted Performance: Medium

13.Conclusion

The Student Performance Prediction System demonstrates how Machine Learning can be used to analyze student-related data and predict performance categories.

The project included data cleaning, exploratory data analysis, feature selection, model training, and model evaluation using a Decision Tree Classifier.

The system was also used to make a prediction for a new student based on their academic and lifestyle-related information.

This project helped in understanding the basic workflow of a Machine Learning project using Python and Scikit-learn.

14.Project Limitations

- The dataset used in this project is small and was created for demonstration purposes.
- The model's accuracy is based on a limited testing dataset.
- The prediction should not be considered a reliable real-world assessment of student performance.

- A larger real-world dataset could be used in the future to improve the model.

15.Future improvements

The project can be improved in the future by:

- Using a larger real-world student dataset.
- Testing different Machine Learning algorithms.
- Comparing the performance of multiple models.
- Improving the prediction accuracy with more data.
- Creating a simple web interface for users.
- Allowing users to enter student details and receive predictions automatically.

16.Project Files

The project contains the following files:

- **Student_Performance_Prediction.ipynb** — Jupyter Notebook containing the complete Python code and analysis.
- **student_performance_final.csv** — Final dataset used for the project.
- **student_performance_model.pkl** — Saved Decision Tree Machine Learning model.
- **Project Report** — Documentation of the project, methodology, results, and conclusions.

